

Cartup Business Intelligence Case Study

Data Analyst – Business Intelligence Hiring Assessment

Background

Cartup is an e-commerce company that connects customers with sellers through its online platform while managing a large part of the operational and delivery process internally.

When a customer places an order on the platform, the order moves through several stages before reaching the customer. These stages include seller processing, warehouse and sorting hub operations, last-mile delivery, and sometimes return handling.

Each of these steps generates operational and transactional data. The Business Intelligence (BI) team at Cartup analyzes this data to understand how the business is performing and to identify opportunities for improvement across operations, commercial performance, and financial outcomes.

Recently, different teams across the company have raised questions about operational efficiency, seller performance, delivery outcomes, campaign effectiveness, and financial impact.

To investigate these questions, you have been given access to several datasets from Cartup's internal systems.

Your task is to analyze the data and identify insights that could help the company make better decisions.

Dataset Overview

You will work with the following tables.

dimcustomers

Contains customer information including name, gender, contact information, and location.

dimshops

Contains seller information including shop names and seller location.

dimproducts

Contains product details including category, selling price, cost price, and current stock levels.

dimriders

Contains delivery rider information including assigned hub and location.

campaigndays

Contains records of promotional campaigns including campaign dates and discount percentages.

orderjourneydata

This table records the lifecycle of each order from placement to delivery or return, including timestamps for different operational stages.

Scenario

Several teams within Cartup have recently reported operational and business concerns:

- Some sellers appear to take longer than expected to process orders.
- Certain orders remain stuck in operational stages longer than expected.
- Delivery outcomes vary significantly across riders and locations.
- Promotional campaigns increase order volumes, but their broader impact is not fully understood.
- Some products and sellers appear to drive more business value than others.
- Returns and delivery failures may be impacting the company financially.

The Business Intelligence team has asked you to analyze the available data to understand what might be happening inside the business.

Your goal is to use the data to identify patterns, uncover possible issues, and suggest opportunities for improvement.

Tasks

**** You may use SQL or any analytical tool to explore the data. ****

During the presentation round, you will be expected to explain:

- How you approached the analysis
- What patterns or insights you discovered
- What these findings mean for the business
- What recommendations you would make to improve performance

Question 1 — Seller Order Processing Performance (Operations & Commercial)

Cartup has internal expectations regarding how quickly sellers should process orders after they are placed (Ideally 24 hours).

In general, sellers are expected to process orders within a reasonable time after the order enters the system. However, the operations team believes that some sellers frequently take longer than expected to process orders.

At the same time, sellers with very high daily order volumes may require more operational flexibility.

Analyze seller-level order processing behavior and identify sellers whose order handling patterns may require operational attention.

Question 2 — Order Journey and Operational Bottlenecks (Operations)

Orders pass through several operational stages before they are delivered to customers.

The operations team suspects that delays may occur in certain stages of the order journey or in specific operational locations.

Analyze how orders move through the operational pipeline and identify where delays or bottlenecks might be occurring.

Consider how these delays might affect overall delivery performance.

Question 3 — Rider Delivery Performance (Operations)

Delivery riders play a critical role in the final stage of order fulfillment.

However, delivery outcomes appear to vary across different riders.

Analyze rider-level delivery performance to understand how delivery outcomes differ between riders.

Your analysis should help identify performance patterns, workload differences, or other factors that may affect delivery success.

Question 4 — Campaign Impact on Operations (Marketing & Operations)

Cartup regularly runs promotional campaigns that offer discounts to customers.

These campaigns are designed to increase sales and attract more customers, but they may also place additional pressure on operations.

Analyze how order patterns and operational outcomes differ during campaign periods compared to regular days.

Your analysis should help determine whether campaigns have any operational impact on the business.

Question 5 — Commercial Performance of Products and Sellers (Commercial)

The commercial team wants to better understand which products and sellers are driving the business.

Analyze product and seller performance to identify key drivers of order volume and revenue.

Your analysis should highlight patterns related to product categories, seller performance, or geographic markets that may help the company improve its commercial strategy.

Question 6 — Financial Impact of Returns and Order Outcomes (Finance)

In e-commerce businesses, order returns and failed deliveries can significantly impact financial performance.

Cartup wants to understand how different order outcomes affect the company financially.

Using the available pricing and cost data, analyze the financial impact of:

- Successfully delivered orders
- Orders that are returned or rejected
- Orders that fail delivery

Your analysis should estimate how much revenue is generated from successful orders and how much value may be lost due to returns or unsuccessful deliveries.

The goal is to understand the financial implications of order outcomes and identify areas where the company could potentially reduce losses.

Deliverables

Each candidate should prepare a short presentation that includes:

- Overview of the analysis performed
- Key insights discovered in the data
- Interpretation of what the findings mean for the business
- Recommendations for Cartup

Candidates are encouraged to explore the data beyond the questions provided if they discover additional interesting insights.